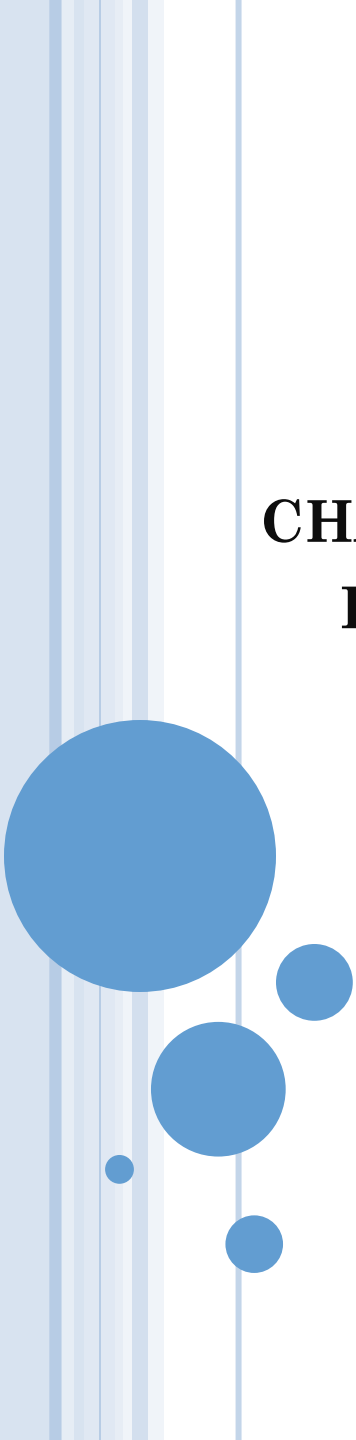




TÍTULO DO ARTIGO: THE EVOLUTION, CHALLENGES, AND FUTURE OF KNOWLEDGE REPRESENTATION IN PRODUCT DESIGN SYSTEM

Chandrasegaran, S. K., Ramani, K., Sriram, R. D., Horváth, I., Bernard, A., Harik, R. F., & Gao, W. (2013). The evolution, challenges, and future of knowledge representation in product design systems. Computer-Aided Design, 45(2), 204–228.

1 ANÁLISE DA ESTRUTURA

ABSTRACT

1 Introduction

2 What is knowledge in product design?

2.1 Classification of knowledge

2.2 Classification of knowledge representation

3 Research in product design: the knowledge perspective

3.1 Design processes view of knowledge

3.1.1 Conceptual design

3.1.2 Function to form mapping and representation

3.1.3 Modeling and simulation

3.1.4 Ontology and the semantic web

3.2 Integration of interacting knowledge systems

3.2.1 Systems architecting and requirements specifications

**Desenvolvimento
do trabalho**



1 ANÁLISE DA ESTRUTURA

3.2.2 *Collaborative engineering*

3.2.3 *Design rationale encoding and use*

4 Computational support tools in design

4.1 *Process view of computer support tools*

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4.1.2 *Design decision making*

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4.2.1 *Collaborative design*

4.2.2 *Systems integration support*

4.2.3 *Product Lifecycle Management (PLM)*

4.2.4 *Knowledge management and mass customization*

**Desenvolvimento
do trabalho**



1 ANÁLISE DA ESTRUTURA

4.3 Computational support tools – an industry view

5 Discussion and future trends



**Infere a partir dos
dados obtidos**

Acknowledgments

References



2 ANÁLISE DO RESUMO

ABSTRACT

Product design is a highly involved, often ill-defined, complex and iterative process, and the needs and specifications of the required artifact get more refined only as the design process moves toward its goal. An effective computer support tool that helps the designer make better-informed decisions requires efficient knowledge representation schemes. In today's world, there is a virtual explosion in the amount of raw data available to the designer, and knowledge representation is critical in order to sift through this data and make sense of it. In addition, the need to stay competitive has shrunk product development time through the use of simultaneous and collaborative design processes, which depend on effective transfer of knowledge between teams. Finally, the awareness that decisions made early in the design process have a higher impact in terms of energy, cost, and sustainability, has resulted in the need to project knowledge typically required in the later stages of design to the earlier stages.



2 ANÁLISE DO RESUMO

Método empregado

Objetivos

Research in design rationale systems, product families, systems engineering, and ontology engineering has sought to capture knowledge from earlier product design decisions, from the breakdown of product functions and associated physical features, and from customer requirements and feedback reports. VR (Virtual reality) systems and multidisciplinary modeling have enabled the simulation of scenarios in the manufacture, assembly, and use of the product. This has helped capture vital knowledge from these stages of the product life and use it in design validation and testing. While there have been considerable and significant developments in knowledge capture and representation in product design, it is useful to sometimes review our position in the area, study the evolution of research in product design, and from past and current trends, try and foresee future developments. The goal of this paper is thus to review both our understanding of the field and the support tools that exist for the purpose, and identify the trends and possible directions research can evolve in the future.

Apresentação dos resultados

2 ANÁLISE DO RESUMO

Keywords: Knowledge representation, Knowledge capture, Knowledge management, Product design, Computational tools, Ontology, Systems engineering, Design rationale, Multidisciplinary modeling, Virtual reality, Collaborative engineering, Simulation



3 ANÁLISE DA INTRODUÇÃO

Objetivo
contribuição e
da
proposta

In this paper, we attempt a review of the product design process from a knowledge capture and representation perspective, keeping in mind the designer, the design process, and the current demands of the market on the product design industry.

The paper structure, indicated in Fig. 1, is as follows: Section 2 discusses the perception of knowledge, its types and its representations with respect to product design. Section 3 looks at knowledge processing in design from two perspectives: (1) individual applications (e.g. conceptual design, modeling and analysis, etc.), and (2) integrated system view, especially collaborative design. Section 4 discusses the computational tools that are being developed for design and how these tools handle knowledge representation. Finally Section 5 discusses future trends in research in this area.

Contextualiza
o trabalho

Estrutura e
organizaçã
o do artigo



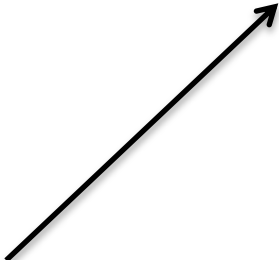
4 ANÁLISE DO DESENVOLVIMENTO

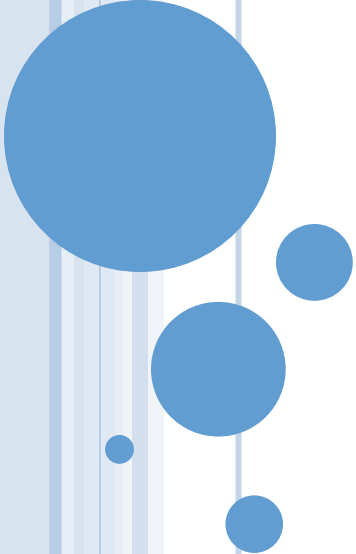
	Método	Finalidade	Característica
Types of knowledge	Formal/ Tacit	Formal knowledge creates the intellectual platform necessary to build and manufacture a product. Tacit knowledge is necessary to create new value in a product.	Tacit knowledge is that the individual has acquired throughout life, which is in people's minds. Formal knowledge is that formal and systematic.
	Product/ Process	Product knowledge, various kinds of relationships between parts and assemblies, geometry, functions, behavior, various constraints associated with products, and design rationale. Process knowledge is mainly concerned with activities associated with the manufacturing floor.	Product knowledge includes various pieces of information and knowledge associated with the evolution of a product throughout its lifecycle. Process knowledge can be classified into design process knowledge, manufacturing process knowledge, and business process knowledge.
	Compiled/ Dynamic	Dynamics knowledge can be classified into qualitative knowledge and quantitative knowledge.	Compiled knowledge is essentially knowledge gained from experience that can be compiled into rules, plans or scripts, cases of previously solved problems, etc. Dynamic knowledge encodes knowledge that can be used to generate additional knowledge structures, not covered by compiled knowledge.
Types of knowledge representation	Definition	Bringing knowledge forward and making it explicit is one of the key roles of knowledge representation.	Pictorial, Symbolic, Linguistic, Virtual, Algorithmic.

5 ANÁLISE DA CONCLUSÃO

We see a strong trend towards the true democratization of product conception and realization, influenced by two main factors: (1) the development of design and programming frameworks that are becoming more accessible and intuitive to use and learn, and (2) the developments in manufacturing and supply chain practices towards a more scale-free framework.

**Inferência a partir dos
dados obtidos**





TÍTULO DO ARTIGO: BASE FRAME CALIBRATION FOR COORDINATED INDUSTRIAL ROBOTS

Gan, Y., & Dai, X. (2011). Base frame calibration for coordinated industrial robots. *Robotics and Autonomous Systems*, 59(7–8), 563–570.

1 ANÁLISE DA ESTRUTURA

ABSTRACT

1 Introduction

2 Calibration procedure for coordinated robots

2.1 Some fundamental relations

2.2 *Calibration of the transformation matrix*

3 Orthonormalization of the rotation matrix

3.1 *Quaternion and the rotation matrix*

3.2 *Optimal approximation of the rotation matrix*

4 Noise analysis summary

5 Simulation and experiment

5.1 *Numerical simulation*

5.2 *Experiment result*

6 Conclusion

Appendix. Transformation between Euler angle and the rotation matrix

References

Procedimento
metodológico

Resultados obtidos

Inferir a partir dos resultados
obtidos



2 ANÁLISE DO RESUMO

ABSTRACT

Objetivo do trabalho

Método de trabalho

To solve the problem of base frame calibration for coordinated multi-robot system, a new method is proposed in this paper. It is carried out through a series of “handclasp” manipulations between two coordinated robots, then a preliminary result can be reached by the calibrating equation. After that, in order to make sure that the calibrated rotation matrix is orthonormal, an optimal estimation of the relative rotation between the base frames of coordinated manipulators is solved out under the criterion of optimal Frobenius norm approximation. By the quaternion representation for rotation matrix and the Lagrange Multiplier method, an orthonormal matrix can be reached which is just the unknown calibrating result for base frames of the coordinated robots. Simulation and experiment results have verified the validity and effectiveness of the proposed method.

Keywords: Base frame calibration, Multi-robot system, Quaternion, Rotation matrix orthonormalization, Coordinated robot

Conclusão a partir dos resultados obtidos



Descreve a problemática e
faz uma contextualização
histórica

3 ANÁLISE DA INTRODUÇÃO

Base frame calibration, which is to determine the relative translation and rotation between base frames of the coordinated robots, is a challenging and fundamental problem for coordinated multirobot system [1]. A direct measurement is inaccessible because the origins the robot base frames are out of reach. Calibrations for an individual robot have already been investigated extensively and many effective methods have been developed [2–7]. Unfortunately, few studies have been made on this problem.

In this paper, a new kind of base frame calibration method is to be discussed. Unlike the above-mentioned calibration method [8–11], it needs no external calibration apparatus or elaborate setups. Inspired by the calibration method in [2,5,7,11], this method uses only a series of “handclasp” manipulations and their corresponding joint information to calibrate the transformation relation of the base frame for the coordinated robots. The great advantage of this method is the easy operational procedure and simple calibrating condition, which makes it quite feasible for use in manufacturing field.

Apresenta o objetivo do trabalho e as
vantagens da proposta em relação a
outras já existentes

3 ANÁLISE DA INTRODUÇÃO

Details of the calibration method will be discussed in Section 2. Section 3 presents an orthonormalization method for the rotational part of the calibrated result since noise exists in the calibration procedure. Noise analysis is summarized in Section 4. Simulation and experiment results are described in Section 5 and the paper ends with concluding remarks in Section 6.

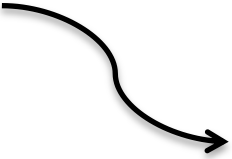
**Apresenta a estrutura do
artigo**



4 ANÁLISE DO MÉTODO

Apesar do método empregado não está explicitamente descrito, o trabalho segue a seguinte metodologia:

2. Calibration procedure for coordinated robots

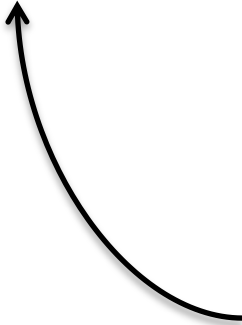


Aqui são introduzidos os conceitos de relações de transformação de juntas de base para dois robôs coordenados, e é mostrado a calibração da matriz de transformação



4 ANÁLISE DO MÉTODO

3. Orthonormalization of the rotation matrix



Nesta seção do procedimento metodológico, é mostrada a necessidade da matriz de rotação entre duas juntas cartesianas ser normalizada e ortogonal. Portanto, inicialmente o problema da matriz de rotação é investigada pela abordagem com quaternions, e, depois, deduz-se uma aproximação ótima da matriz de rotação



4 ANÁLISE DOS RESULTADOS

4. Noise analysis summary

With the method proposed here, noise from three aspects will make the calibration result degenerated.

(1) Inaccuracy in mechanism parameters.

The robot tool and link parameters are used in the calibration, which is provided by the robot and end-effector manufacturers. However the robot and its tools cannot be produced without inaccuracy, these mechanism inaccuracy will be involved in the calibration result.

(2) Noise in robot joint position sensors.

Noise is unavoidable with the use of robot joint position sensor, and these noised data will be applied in the calibration procedure, which will certainly make the calibration result inaccurate.

Assim como a metodologia, os resultados não estão em uma seção explícita. Os resultados iniciais estão relacionados à análise de ruído de três fatores, que alteram o resultado da calibração para o método de calibração proposto no trabalho

4 ANÁLISE DOS RESULTADOS

The above two inaccuracy sources cannot be controlled by the operator during the calibration process, which are determined by the robot manufacturers. Fortunately, these inaccuracy is relatively small. As long as no defection exists in the coordinated robots, these two inaccuracy sources can be omitted.

(3) Inaccuracy in handclasp operation.

During the handclasp operation between two coordinated robots, it is expected to move both the robot TCP points to the same place. Of course, that is just an ideal state, and inaccuracy exists in this operation. So, in order to make the calibration result more accurate, operators should be as careful as possible and make the two TCP points to one same place more accurately.

For reducing this inaccuracy, a measuring tip is recommended to replace the usual robot end-effectors, which is also mentioned in the end of Section 2. At the same time, the four noncoplanar points should be selected as far apart as possible. A distance no less than 100 mm between these selected points is strongly recommended, which is an empirical value in our experiment testbed.

↑ Soluções propostas pelos autores para
redução da imprecisão gerada pelos fatores
destacados em vermelho



4 ANÁLISE DOS RESULTADOS

5. Simulation and experiment

Simulação numérica para comprovar a eficácia do método.

To illustrate the validity and effectiveness of our proposed calibration method, a numerical simulation is carried out for two coordinated Puma560 robots.

Os resultados obtidos mostram que os erros de calibração são pequenos, atendendo às exigências de precisão para as tarefas a serem desempenhadas pelos robôs

Now, calibration error for the rotation matrix can be calculated as follow,

$$e_{\text{rot}} = \left\| \text{Euler}({}^b R_{bj}^{\text{tru}}) - \text{Euler}({}^b \tilde{R}_{bj}) \right\|_{\infty} = 0.0055. \quad (46)$$

The error e_{rot} with a unit of angle (rad), has a definite meaning that represents the maximum error between Euler forms of the calibrated rotation matrix and its true value. The calibration error for the translation vector can be calculated as,

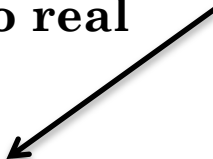
$$e_{\text{tran}} = \left\| {}^b T_{bj}^{\text{tru}} - {}^b \tilde{T}_{bj} \right\|_{\infty} = 0.0036. \quad (47)$$

The error e_{tran} has a unit of length (m), whose meaning represents the maximum error between calibrated translation vector and its true value.

Up to now, we have completed the base frame calibration procedure for our simulation case. Eqs. (46) and (47) indicate that the calibration errors are relatively small, which can meet the precision requirement for most rough robot tasks, such as arc welding or material transporting.

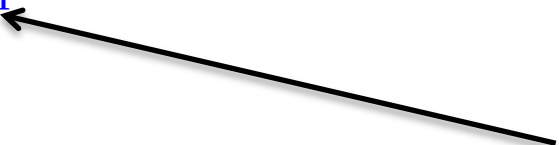
4 ANÁLISE DOS RESULTADOS

Teste do método de
calibração em
experimento real



Execute the calibration procedure on this real experiment testbed, the following calibration results are acquired in Table 1.

From Table 1, it is clear that precision of our proposed method is no more than 0.011 m in position calibration and 0.026 rad in orientation calibration. This precision result is quite satisfactory that can meet the requirement for most rough robot tasks. Now, considering both the simulation and experiment results, it can be concluded that the calibration procedure proposed in this paper is quite effective to solve the problem of base frame calibration for two coordinated industrial robots.



Os resultados de precisão, obtidos com o experimento, permitem verificar que os mesmos são satisfatórios para as atividades áspersas a serem desempenhadas pelo robô

Além disso, os resultados numéricos e experimentais comprovam a eficácia do método de calibração proposto neste trabalho



5 ANÁLISE DA CONCLUSÃO

Simulation and experiment testbed of base frame calibration with two coordinated industrial robots is discussed. The calibration result is quite satisfactory which asserts the validity and effectiveness of the proposed method. Defect of our calibration method may lie in that handclasp manipulation between the coordinated robots is driven by human operator. If the TCP point of each coordinated robot cannot be exactly driven to one same point, error will expand in the calibrated result. As long as the human operator drive the robot carefully and make the handclasp manipulation as accurate as possible, the calibration error can be reduced to relatively small.

Inferências a partir dos
resultados obtidos

